



EDU TERIA

72nd BPSC

Prelims Online Batch

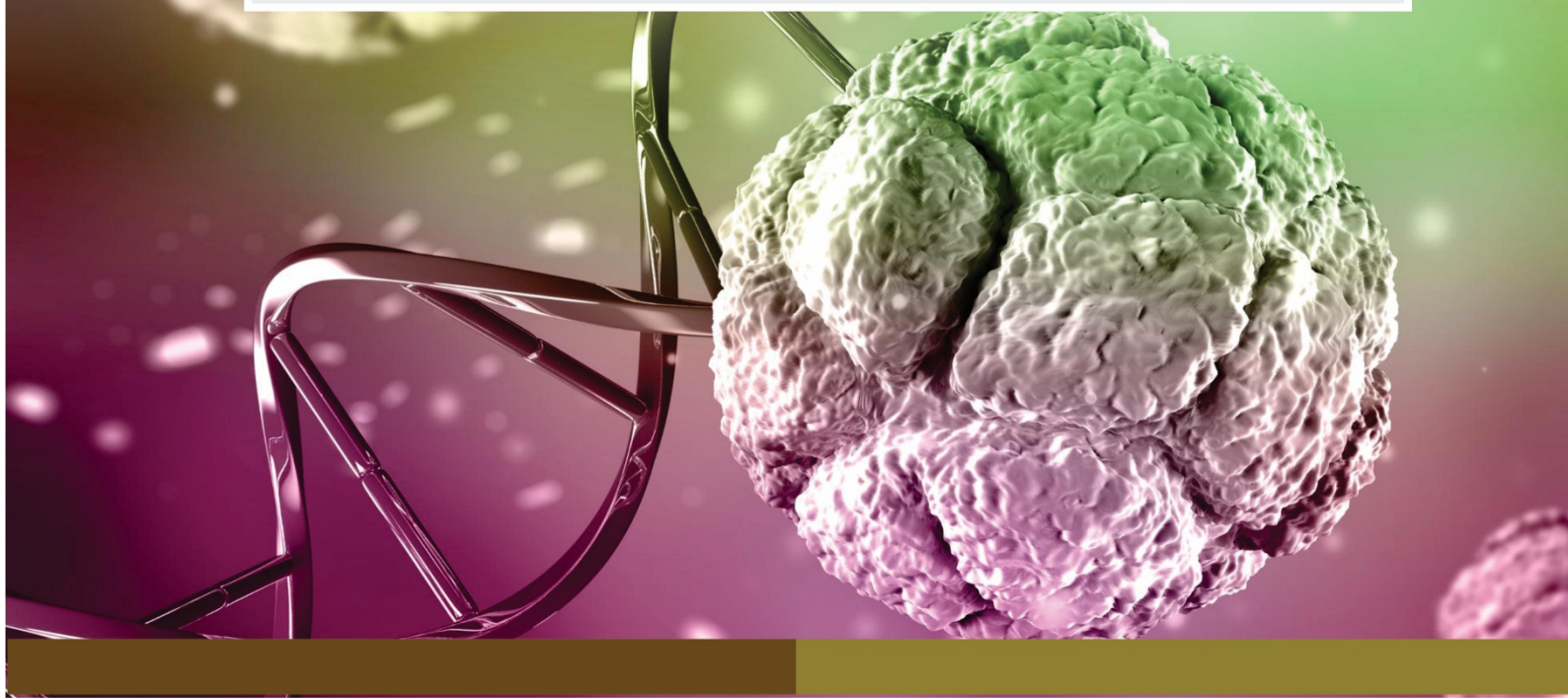
BIOLOGY

Topic wise
Notes

LECTURE

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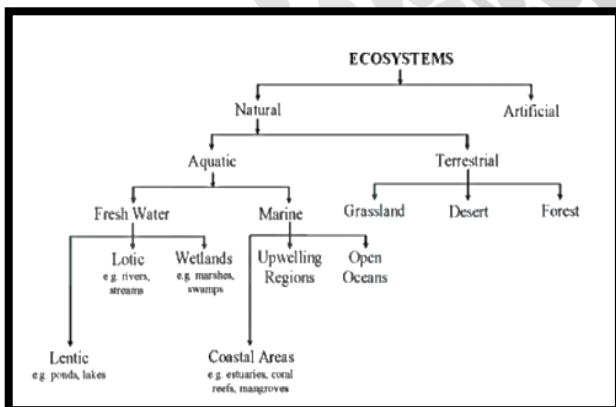
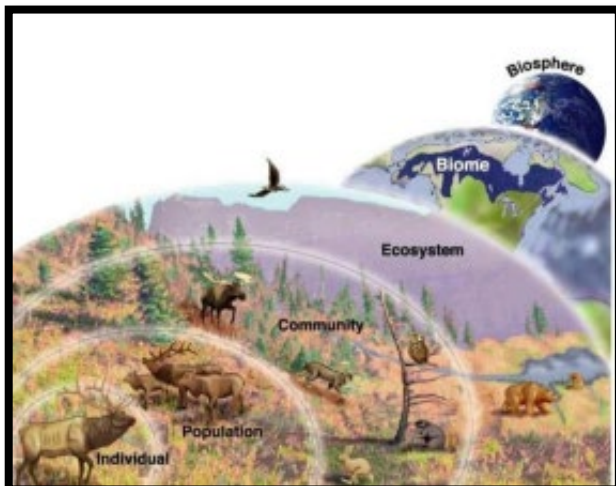
Ecology



Ecology

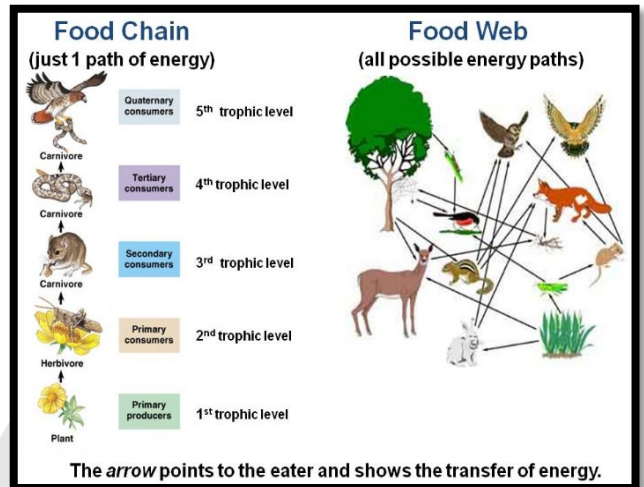
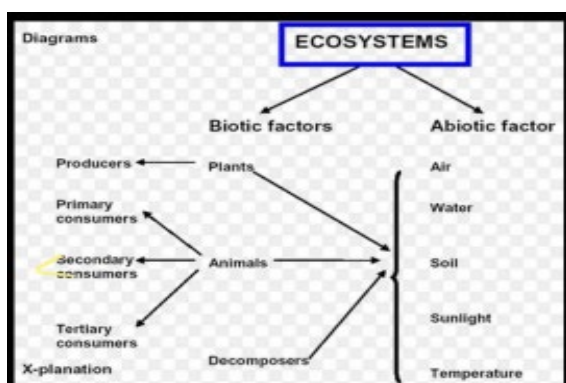
Ecology

Branch of science which deals with the study of interaction between organism and its surrounding environment known as Ecology.



- Ecology word was termed by— Ernest Haeckel
- Father of Ecology— Reiter

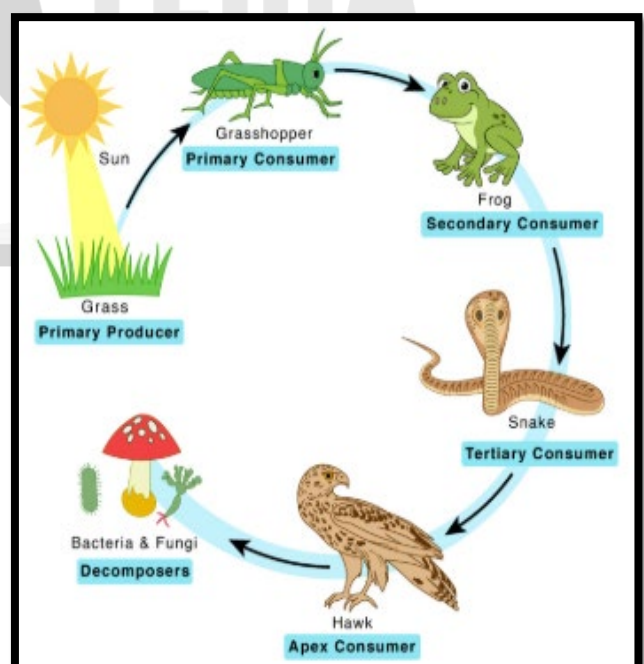
Components of Ecology:



Biotic Component	
Producer	Consumer
Generally plants	Other than plants
Perform photosynthesis or chemosynthesis	Heterotrophs

Food Chain

Food Chain is a linear sequence of organisms through which food, energy, and nutrients pass as one organism eats another in an ecosystem. It shows the transfer of energy from producers to consumers and finally to decomposers.

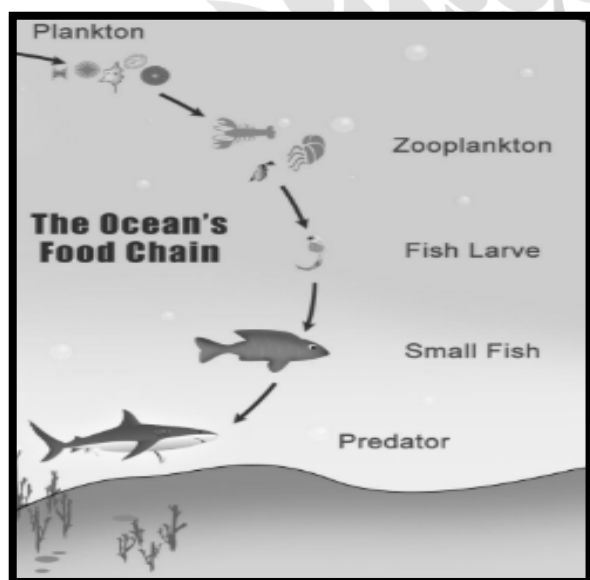


Food Chain			
Grazing food chain		Parasitic food chain	Detritus food chain
Terrestrial food chain	Aquatic food chain	Depends upon parasite	Starts with detritivores (Bacteria/Fungus)

Aquatic Food Chain

An Aquatic Food Chain is the sequence of organisms in a water ecosystem through which food and energy are transferred from one organism to another. It begins with aquatic producers like phytoplankton and ends with top consumers.

Example: Phytoplankton , Zooplankton, Small Fish, Large Fish, Human/Bird



Ecological Pyramids

Ecological Pyramid are graphical representations that show the relationship between different trophic levels in an ecosystem in terms of number, biomass, or energy. Producers form the base of the pyramid, while consumers occupy the upper levels. Ecological pyramids are of three types: pyramid of numbers, pyramid of biomass, and pyramid of energy.

Types of Ecological Pyramids

1. Pyramid of Numbers

It represents the number of organisms present at each trophic level.

Example: Grass, Grasshopper, Frog , Snake ,Hawk

Features

Usually upright in grassland ecosystems.
May be inverted in tree ecosystems because one large tree supports many herbivores.

Types

Upright Pyramid of Numbers

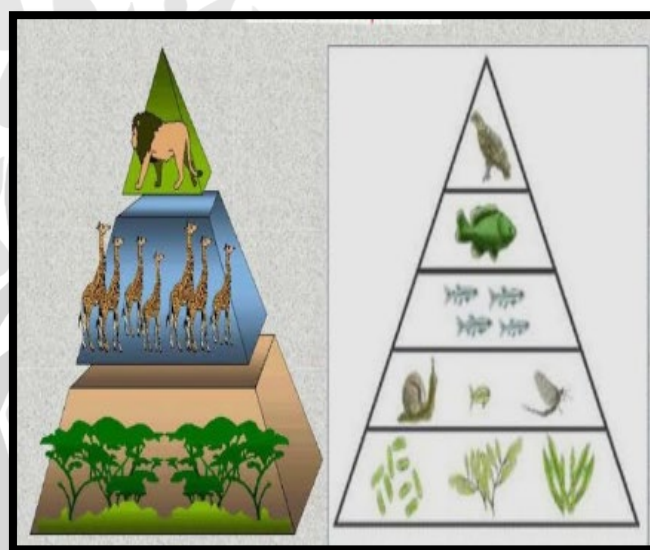
Many producers support fewer herbivores and still fewer carnivores.

Example: Grassland ecosystem

Inverted Pyramid of Numbers

One producer supports many consumers.

Example: Tree ecosystem and parasitic food chain



2. Pyramid of Biomass

It represents the total dry weight or living matter present at each trophic level.

Features

Biomass is measured in gram/m² or kg/m².
Indicates the standing crop of organisms.

Types

Upright Pyramid of Biomass Found in terrestrial ecosystems like forests and grasslands.

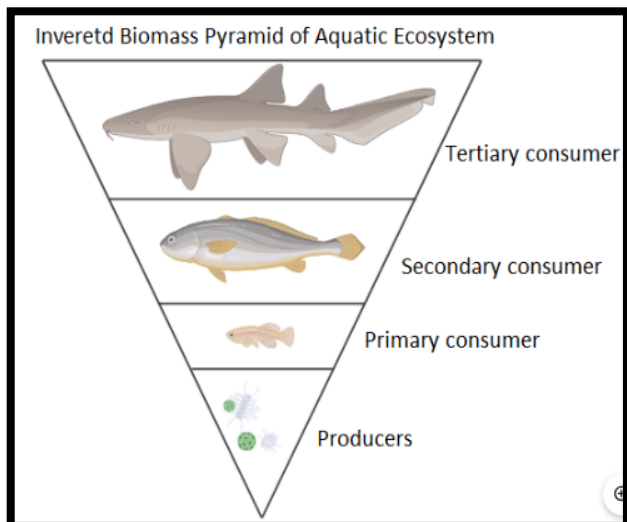
Example: Trees , Deer. Tiger

Inverted Pyramid of Biomass

Found in aquatic ecosystems because phytoplankton have less biomass than zooplankton and fishes at a given time.

Example: Phytoplankton, Zooplankton , Fish





3. Pyramid of Energy

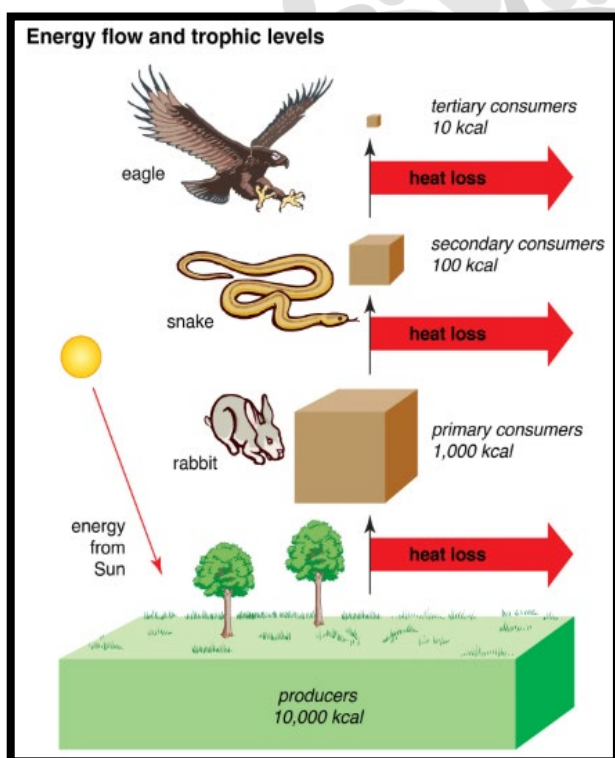
It represents the flow of energy at different trophic levels.

Features

- Always upright.
- Energy decreases at every trophic level according to the 10% law.

Measured in $\text{kcal/m}^2/\text{year}$ or $\text{joules/m}^2/\text{year}$.

Example: Plants, Herbivores, Carnivores



Types of Species Interaction

Species interactions are the relationships between different species living in the same ecosystem. These interactions may be beneficial, harmful, or neutral for the organisms involved.

1. Mutualism

Both species are benefited from the interaction.

Examples : Lichen, Association between algae and fungi. Bees and flowering plants, Bees get nectar; plants get pollination. Rhizobium bacteria and leguminous plants, Bacteria get shelter; plants receive nitrogen.

Importance

Helps in nutrition, reproduction, and survival.

2. Commensalism

One species benefits while the other is neither harmed nor benefited.

Examples : Orchid growing on mango tree. Remora fish attached to shark. Cattle egret feeding near grazing cattle.

Features

One partner gains food, shelter, or transport.

3. Parasitism

One organism (parasite) benefits while the other (host) is harmed.

Types

Ectoparasites

- Live outside the host body. Example: Lice, ticks

Endoparasites

- Live inside the host body. Example: Tapeworm, Plasmodium

Examples: Cuscuta on plants, Mosquito on humans

Harmful Effects

Causes diseases and weakness in host organisms.

4. Predation

One organism (predator) kills and eats another organism (prey).

Examples : Lion and deer, Snake and frog, Frog and insects.

Importance

Maintains ecological balance.

Controls population size.

5. Competition

Two species compete for the same limited resources like food, water, light, or shelter.

Types

- Intraspecific Competition





- Occurs between members of the same species.
- Interspecific Competition

Occurs between different species.

Examples : Weeds and crops competing for nutrients., Lions and hyenas competing for prey.

Effects

Reduces growth and survival rate.

6. Amensalism

One species is harmed while the other remains unaffected.

Examples : Penicillium fungus secretes antibiotics that kill bacteria. Large trees preventing growth of small plants by shading.

7. Neutralism

Neither species affects the other.

Example : Deer and squirrel living in the same forest without direct interaction.

Special Types of Interactions

I. Symbiosis

A close and long-term association between two species. It usually refers to mutualism.

Example: Lichen

II. Protocooperation

Both species benefit, but the association is not compulsory.

Example: Hermit crab and sea anemone

